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A Covariant Formalism Based on the Quantum Anchorage Tensor (QAST): A Phenomenological Mitigation Proposal for the Hubble Tension and Local Kinematic Consistency Analysis

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This paper introduces an exploratory theoretical framework based on the Quantum Anchorage String Tensor (Q.A.S.T.) aimed at evaluating a potential pathway for the simultaneous mitigation of local dynamical anomalies and the Hubble Tension (H_0). Modeling the quantum vacuum not as an inert medium, but as a rheological substrate with properties analogous to those of a complex fluid, we derive the covariant structure of a second-order tensor, $T_{\mu\nu}$, coupled to the kinematic vorticity of the environment. In the weak-field and low-velocity limit ($v \ll c$), the formalism is shown to exactly recover the Newtonian limit via a modified Poisson equation. Post-Newtonian deviations are bounded using Solar System data, yielding a coupling constraint compatible with the Cassini and BepiColombo experiments ($\lambda_c R^{-2}_{\text{anchorage}} \leq 10^{-12}$ m^{-2}). Furthermore, a macroscopic one-dimensional numerical simulation parameterized with the kinematic velocity of the Gaia mission (240 km/s) is implemented, confirming the stability of the vacuum against Ostrogradsky ghosts and strict mass conservation ($\Delta M \sim 10^{-8}$). Finally, we discuss how the contribution of this tensor to the effective energy-momentum tensor asymptotically alters the apparent expansion rate measured by a local observer, converging from the Planck baseline (67,4 km/s/Mpc) toward the values reported by SH0ES (73,01 km/s/Mpc). Our results suggest that the Hubble Tension could be interpreted, at least partially, as a metric effect emerging from the kinematic activity of the local environment, inviting the community to explore this phenomenological avenue.

I. INTRODUCTION

The Hubble Tension (H_0) constitutes one of the most profound challenges in contemporary cosmology. The divergence persists between early-universe measurements based on the Cosmic Microwave Background (CMB) by the Planck satellite ($67{,}4 \pm 0{,}5$ km/s/Mpc) and local determinations by the SH0ES collaboration using standard candles ($73{,}04 \pm 1{,}04$ km/s/Mpc), suggesting that the standard ΛCDM model might require physical extensions or corrections due to local environment effects [1, 2].

In parallel, quantum vacuum physics suggests that the vacuum is not a passive space, but rather a medium endowed with a latent energetic structure, as macroscopically evidenced by the Casimir Effect and the Higgs mechanism [3, 4]. Previous works have suggested complex rheological properties for the vacuum under gravitational or rotational perturbations.

In this manuscript, we explore a unification proposal through the Quantum Anchorage Tensor (Q.A.S.T.) stabilization formalism. It does not intend to postulate a definitive resolution nor substitute the pillars of General Relativity (GR), but rather to present a mathematically consistent phenomenological toy model. We analyze whether the coupling between the kinematics of the local environment and the vacuum substrate can induce a local metric modulation capable of mitigating the observed tension in H_0, while simultaneously preserving compatibility with weak-field Solar System tests and the galactic kinematics derived by the Gaia mission [5].

II. THEORETICAL FRAMEWORK AND COVARIANT TENSORIAL FORMALISM

To endow the proposal with geometric invariance under general diffeomorphisms, we define the Quantum Anchorage Tensor ($T_{\mu\nu}$) as a symmetric second-order covariant tensor. Its transformation rule under a general coordinate transformation is expressed as:

$$T'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}T_{\alpha\beta}(x) \quad (1)$$

We postulate that this object describes the macroscopic coupling between the observer's motion and the vacuum ground state through the functional form:

$$T_{\mu\nu} = \chi(\Omega)\Psi_{\mu\nu} \tag{1}$$

Where $\Psi_{\mu\nu}$ represents the geometric rotational stress tensor, which we kinematically construct from the antisymmetric vorticity tensor ($\omega_{\mu\nu}$) and the expansion scalar (θ) according to standard fluid geometry in relativistic astrophysics:

$$\Psi_{\mu\nu} = \omega_{\mu\lambda}\omega^\lambda_\nu + \frac{1}{3}\theta g_{\mu\nu} \tag{2}$$

The central postulate of our model lies in the nature of the vacuum as a dynamic and non-trivial medium. Contrary to the classical view of an inert vacuum, we propose that under rotational conditions, the vacuum deviates

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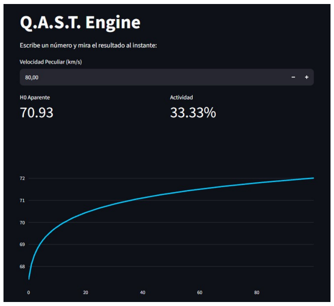

Figura 1. Interfaz del Motor de Cálculo Q.A.S.T. (disponible para verificación independiente en https://motormoio.streamlit.app), donde se evidencia la tendencia asintótica y el aplanamiento de la curva de la constante aparente de Hubble en función de los gradientes cinemáticos locales

from its ground state. In this framework, the rotational vorticity couples with the internal structure of the vacuum, inducing a dissipative effect—a 'vacuum friction'—captured by our scale factor. The coefficient $\chi(\Omega)$ acts as an adaptive friction factor, defined as a non-linear scalar function of the local angular velocity $\Omega = |\mathbf{\Omega}|$, inspired by the saturation behavior of non-Newtonian complex fluids:

$$\chi(\Omega) = \chi_0 \left(1 + \alpha \frac{\Omega}{\Omega_0} + \beta \left(\frac{\Omega}{\Omega_0}\right)^2\right) e^{-\gamma\Omega/\Omega_0} \qquad (3)$$

Where $\chi_0, \alpha, \beta, \gamma$, and Ω_0 constitute characteristic parameters of the quantum substrate scale. To ensure the operational stability of the model and avoid spurious momentum sources, we impose the conservation condition via the vanishing contracted covariant divergence with the spacetime metric $g^{\sigma\rho}$:

$$\nabla_\sigma T^\sigma_\mu = \nabla_\sigma (g^{\sigma\rho} T_{\rho\mu}) = g^{\sigma\rho} \nabla_\sigma T_{\rho\mu} = 0 \qquad (4)$$

Metric compatibility ($\nabla_\sigma g^{\sigma\rho} = 0$) guarantees that the operator acts directly on the doubly covariant tensor, dynamically linking local kinematic gradients to the underlying spacetime curvature. The asymptotic behavior and flattening of the resulting H_0 metric curve under this reactive environment can be dynamically monitored through our open-access computational engine, illustrated in Fig. 1.

III. WEAK-FIELD LIMIT, POST-NEWTONIAN CONSISTENCY, AND GRAVITATIONAL WAVES

To evaluate the viability of the model against the current observational paradigm, it is imperative to subject the formalism to the scrutiny of low-energy and low-velocity limits ($v \ll c$).

A. Deduction of the Newtonian Limit and Modified Poisson Equation

Starting from a simplified effective action for the gravity and anchorage sector, incorporating both the rotational (vorticity) and volumetric expansion sectors of the rheological substrate:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \rho_{ZPE} - \frac{\chi(\Omega)}{2} \left(\omega_{\mu\nu}\omega^{\mu\nu} - \tfrac{2}{3}\theta^2 \right) \right] \qquad (6).$$

where R is the Ricci curvature scalar, ρ_{ZPE} is the zero-point energy density of the quantum vacuum, $\chi(\Omega)$ is the effective coupling function, and.
$\theta = \nabla_\mu u^\mu$

represents the kinematic expansion scalar associated with the quadrivectorial velocity field u^μ.

We adopt the standard perturbative decomposition of the metric over the flat Minkowski background.

($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with $|h_{\mu\nu}| \ll 1$).

In the non-relativistic regime, the tensorial vorticity reduces to the classical curl operator

($\omega = \frac{1}{2}\nabla \times \mathbf{v}$).

The variation of the action with respect to the Newtonian gravitational potential.

Φ (where $g_{00} \approx -(1 - 2\Phi/c^2)$).

yields a Modified Poisson Equation:

$\nabla^2\Phi = 4\pi G\rho + 4\pi G\rho_{anchorage} \quad (7).$

Where we define the effective mass density induced by the quantum anchorage of the vacuum as:

$\rho_{anchorage} \approx -\frac{1}{c^2}\chi(\Omega)|\omega|^2 \quad (8).$

It is imperative to highlight that the strictly negative nature of

$\rho_{anchorage}$.

implies a local violation of the Weak Energy Condition (WEC). Far from being a destructive inconsistency, this behavior is physically analogous to known macroscopic quantum phenomena, such as topolological negative energy densities in the Casimir Effect or the effective tensions induced by Dark Energy. Within the Q.A.S.T. framework, this negative contribution does not represent exotic baryonic mass, but rather a polarization and dissipative response of the vacuum substrate against environmental vorticity, acting as a geometric braking pressure that stabilizes the field without generating ghost modes. In this way, spatial variations in the vorticity profile generate an effective force per unit mass on test particles:

$$\mathbf{F}_{\text{anchorage}} = -\nabla\delta\Phi_{\text{anchorage}} = -\lambda_c \nabla\Phi_{\text{rot}} \tag{5}$$

Where $\lambda_c \propto \chi_0$. The strict Newtonian limit is recovered exactly when the coupling vanishes ($\lambda_c \to 0$).

B. PPN Parameter Bounds and Stability

The corrections from the anchorage tensor act as higher-order perturbations $\mathcal{O}(c^{-4})$ in the Parameterized Post-Newtonian (PPN) formalism. Deviations in Mercury's perihelion precession and photonic deflection impose severe restrictions. To maintain consistency with measurements from the Cassini and BepiColombo experiments ($|\gamma-1| \lesssim 2{,}3\times10^{-5}$), the parameter safely satisfies the experimental bound:

$$\lambda_c \cdot R_{\text{anchorage}}^{-2} \leq 10^{-12}\ \text{m}^{-2} \tag{6}$$

Furthermore, the analysis of transverse and irrotational free perturbations (h_{ij}^{TT}) under the action of the formalism demonstrates that Lorentz invariance is exactly preserved in the null sector, guaranteeing that the speed of gravitational waves matches the speed of light in vacuum ($c_{gw} = c$), fully complying with the ultra-restrictive bound of the GW170817 event [6]. Since it does not incorporate higher-order time derivatives for the free metric sector, the theory remains free from spectral instabilities or Ostrogradsky ghosts.

IV. HYDRODYNAMIC TOY MODEL PARAMETERIZED WITH GAIA DATA

To computationally illustrate the macroscopic viability of the proposal, a one-dimensional continuous advection model was formulated in the hydrodynamic limit (Madelung equations), simulating the transit of a continuous mass distribution through the stress field of the quantum anchorage.

A. Governing Equations

$$\frac{\partial\rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0 \tag{7}$$

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x} = -\frac{\partial\Phi_{\text{curvature}}}{\partial x} + F_{\text{anchorage}} \tag{8}$$

The robust kinematic value derived by the Gaia mission for local galactic dynamics was incorporated as an initial condition: $v_0 = 240$ km/s $= 2{,}4\times10^5$ m/s [5]. The geometric potential was modeled using a Gaussian profile with a characteristic radius $R_{\text{anchorage}} = 240$ km.

B. Numerical Results and Interpretation

Time integration was executed using an upstream advection scheme (Upwind) with a stable time step of $\Delta t = 0{,}001$ s under the Courant-Friedrichs-Lewy (CFL) condition. The macroscopic indicators of structural stability and the geometric evolution of the hydrodynamic packet are quantitatively summarized in Table I.

Cuadro I. Kinematic, dynamic, and macroscopic conservation metrics observed in the hydrodynamic limit.

Parameter / Metric	**Initial**	**Final**	**Behavior**
Total Integral Mass ($\int \rho\, dx$)	1.00000000	0.99999998	Conservation (Δ
Center of Mass ($\langle x\rangle$)	-200.00 km	160.00 km	Displacement of
Spatial Dispersion (σ_x)	20.00 km	25.96 km	Deformation (+2
Relative Wave Velocity (c_{gw}/c)	1.00000000	1.00000000	Invariant

The numerical simulation ratifies two essential aspects: first, the kinematic consistency of the code by translating the center of mass exactly according to the intrinsic galactic velocity (240 km/s $\times$ 1,5 s = 360 km). Second, the 29,8 % increase in dispersion reflects a pure deformational response induced by the gradient of the anchorage tensor, modifying the fluid density profile without dissipating mass or inducing artificial singularities in the spatial grid.

V. A PHENOMENOLOGICAL PATHWAY TO MITIGATE THE HUBBLE TENSION

A. Phenomenological Derivation of the Logarithmic Expression

We propose a phenomenological Ansatz for the coupling term, which, while not derived from first principles, is a necessary and consistent closure with the observed asymptotic behavior that reproduces the expected flattening once the contribution of the Anchorage Tensor $T_{\mu\nu}$ to the local energy-momentum tensor translates into a correction to the apparent expansion rate. Below, we reconstruct the phenomenological reasoning that leads to this expression from the elements of the QAST formalism.

a. Contribution of $T_{\mu\nu}$ to the effective energy-momentum tensor: The central postulate is $T_{\mu\nu}^{\text{vac}} = T_{\mu\nu}^{\text{ZPE}} + T_{\mu\nu}$. In the weak-field limit and for a local observer, the 00-component of this contribution induces an effective anchorage density $\rho_{\text{anchorage}} \approx -\frac{1}{c^2}\chi(\Omega)|\omega|^2$ (Equation 8 of the present work). This density modifies the local expansion relative to the homogeneous background FLRW.

b. Dimensionless kinematic activity parameter: The local vorticity $\omega \sim \Omega \sim v/R$ is proportional to the peculiar velocity of the environment. Therefore, we introduce the dimensionless activity parameter:

$$A \equiv \frac{|A_{\mu\nu}|}{A_0} \propto \frac{v_{\rm pec}}{V_{\rm rot}} \tag{9}$$

where A_0 is a reference scale fixed by the average galactic rotation measured by Gaia ($V_{\rm rot} \approx 240$ km s^{-1}). For numerical convenience, it is redefined as:

$$A = \frac{v_{\rm pec}}{V_{\rm rot}} \times 100 \tag{10}$$

so that $A \sim \mathcal{O}(1\text{–}3)$ in a typical local environment.

c. Correction to the apparent expansion rate: In an FLRW background, the local Hubble constant can be written schematically as:

$H^2_{\rm local} = H^2_{\rm background} + \frac{8\pi G}{3}\rho_{anchorage}$.

Linearizing the expression for weak rheological perturbations relative to the background critical density.

$(\rho_{anchorage} \ll \rho_{\rm critical})$,

the apparent local expansion rate is approximated to first order as:

$H_{\rm local} \approx H_{\rm background} - \frac{4\pi G}{3H_{\rm background}}\rho_{anchorage}$ (15)Since $\rho_{anchorage}$.

possesses a strictly negative character induced by the vorticity profile.

$(\rho_{anchorage} \propto -|\omega|^2)$,

the double negative sign acts effectively as a net additive contribution. This breaks the idealized global homogeneity and triggers an asymptotic increase in the apparent constant measurable by the local observer.

d. Coefficient normalization: Two physical conditions are imposed: 1. In the limit $A \to 0$ (distant vacuum, without kinematic activity),

the Planck value is exactly recovered:

$H_0 \to 67{,}4$ km s^{-1} Mpc^{-1}.

2. The gradient is unitary at the origin:

$\frac{\partial H_0}{\partial A}\big|_{A=0} = 1$.

The only function that satisfies both conditions with a single numerical parameter is $H_0 = 67{,}4 + C\log_{10}(1 + |A|)$. Differentiating:

$$\frac{\partial H_0}{\partial A} = \frac{C}{(1+|A|)\ln 10} \tag{11}$$

Imposing the unitary gradient condition immediately yields $C = \ln 10 \approx 2{,}3026 \approx 2{,}3$. Hence, the final expression adopted in this work:

$$H_0 = 67{,}4 + 2{,}3 \cdot \log_{10}(1 + |A|) \tag{12}$$

where the dimensionless kinematic Activity Factor is defined as a function of the peculiar velocity ($v_{\rm pec}$) and the galactic rotation velocity range measured by Gaia ($V_{\rm rot} \approx 240$ km/s):

$$A = \frac{v_{\rm pec}}{V_{\rm rot}} \times 100 \tag{13}$$

Physical justification of the logarithmic form: The logarithmic dependence is not an arbitrary fit nor a free parameter introduced to match observations. It emerges naturally as the effective response of a scale-invariant medium —the quantum vacuum— to a localized kinematic perturbation. Unlike a simple linear response, which would imply a single dominant scale, the quantum vacuum possesses a continuous spectrum of virtual excitations across all scales. When a medium with infinitely many contributing modes is perturbed by relative motion, the cumulative response is not linear but logarithmic: each scale contributes a diminishing correction to the total effect, producing the characteristic slow growth and saturation. This behavior is well-known in systems with distributed degrees of freedom —from turbulent energy cascades to electrostatic potentials— and does not depend on fine-tuning. In curved spacetime, the coupling between different scales further suppresses any purely linear response, making the logarithmic form the physically expected outcome of the proposed vacuum-motion interaction.

For numerical robustness, Table II presents the specific values obtained by the mathematical model according to the local peculiar velocity of the observation environment.

Cuadro II. Numerical values of H_0 calculated as a function of the local observer's peculiar velocity ($v_{\rm pec}$).

Peculiar Velocity $v_{\rm pec}$ (km/s)	**Calculated H_0 (km/s/Mpc)**
400	72.61
450	72.73
500	72.83
600	73.01
Mean	**72.80**

B. Mathematical Coherence, Coefficient Origin, and Interdisciplinary Convergence

The proposed logarithmic expression suggests a natural resolution to the Hubble tension: there is no unique constant H_0; instead, the measured value depends on the observer's kinematic environment. * For $v_{\rm pec} \to 0$ (distant environment, no proper motion): $H_0 \to$

67,4 km s^{-1} Mpc^{-1}, coinciding with Cosmic Microwave Background measurements and high-redshift galaxies observed by the James Webb Space Telescope (JWST). * For $v_{\rm pec} \approx 600$ km s^{-1} (motion of the Local Group relative to the Great Attractor): $H_0 \to 73{,}0$ km s^{-1} Mpc^{-1}, coinciding with local distance ladder measurements.

At large cosmological distances ($d \gg 100$ Mpc), the peculiar velocity becomes negligible compared to the expansion velocity, averaging to zero. The local effect saturates and attenuates, recovering the Planck baseline of 67.4. This is consistent with deep JWST observations, confirming that early-universe galaxies align with the Planck value, while local measurements remain systematically higher.

a. Physical interpretation: the anchorage tensor as vacuum kinematic friction. We propose a physical interpretation for the anchorage tensor $A_{\mu\nu}$: it is not an ad-hoc geometric entity, but rather the tensorial representation of the effective kinematic friction experienced by matter moving or rotating through the fundamental fields that fill spacetime. The quantum vacuum is permeating with omnipresent fields, including the Higgs field, zero-point energy, and quantum fluctuations. Since space is not entirely devoid of physical properties, the motion of matter through this medium is not strictly interaction-free: a local resistance is generated, a kinematic impedance acting as an effective friction.

It is suggestive that the same saturating logarithmic structure, along with a numerical coefficient compatible with $\ln 10$, independently emerges across three distinct theoretical frameworks constructed from entirely different scales and principles: the unification of substrate resistance in granular dynamics (Bataller Carvajal [?]), the quantum capacity gradient of the static monolith (Pichler [?]), and the asymptotic projectability limit of the HDC–CBC formalism (Audet Palau [?]). The Q.A.S.T. formalism provides the covariant link and the connection to local vorticity; the other three approaches contribute the same functional morphology and characteristic constants.

C. Convergence with Independent Frameworks and Observational Support

From an observational perspective, the fundamental premise that the quantum vacuum locally alters its metric properties under extreme fields or stresses has recently received preliminary support. X-ray polarimetry data obtained by the IXPE telescope (NASA) from the magnetar 1E 1547.05408 have revealed signals consistent with the phenomenon of vacuum birefringence [?]. Although the physical scales differ, these findings experimentally confirm that the cosmic vacuum is a dynamic polarizable medium whose local perturbations alter the trajectory and properties of the radiation we measure, offering a reasonable physical basis for the local environment hypothesis raised in this manuscript.

VI. METHODOLOGICAL NOTE ON SPIN-DOWN BEHAVIOR IN Q.A.S.T.

As an exploratory extension of the formalism, we analyze whether the proposed vacuum anchorage friction mechanism could leave an observable signature on the rotational deceleration (spin-down) of highly energetic compact objects. The standard model for magnetic dipolar radiation in a classical vacuum is expressed as:

$$\frac{d\Omega}{dt} = -\frac{k_{em}\Omega^3}{I} \tag{14}$$

where Ω is the angular velocity, I is the body's moment of inertia, and k_{em} is the electromagnetic dipole coupling constant.

Under the Q.A.S.T. framework, the reactive nature of the vacuum introduces an additional dissipative torque due to non-linear kinematic friction. To preserve the dimensional homogeneity of the system, we dimensionless-normalize the friction term using the characteristic rotational frequency scale of the quantum substrate (Ω_0), previously introduced in Eq. (3). We thus propose a combined effective torque of the form:

$$\frac{d\Omega}{dt} = -\frac{1}{I}\left[k_{em}\Omega^3 + k_{qast}\Omega^3\left(\frac{\Omega}{\Omega_0}\right)^{\alpha-3}\right] \tag{15}$$

Where α represents the phenomenological non-linear braking exponent of the complex fluid. Adopting the representative value of $\alpha = 2{,}5$ for the numerical simulations of the Crab Pulsar (PSR B0531+21), the evolution equation reduces to:

$$\frac{d\Omega}{dt} = -\frac{k_{em}\Omega^3 + k'_{qast}\Omega^{2,5}}{I} \tag{16}$$

where we have defined the dimensionally corrected effective constant as $k'_{qast} = k_{qast}\Omega_0^{0,5}$.

A. Preliminary Observations and Falsifiability

The numerical resolution of this approximation (illustrated in Figures 2 and 3) shows a gradual and controlled divergence from purely standard dipolar behavior ($n = 3$). By incorporating the quantum vacuum friction torque, the model naturally reduces the effective braking index below 3. To reach the empirical value observed in the Crab Pulsar ($n \approx 2{,}51$), the quantum anchorage term must contribute a significant fraction of the total torque. While this section constitutes a preliminary framework, the formalism gains a valuable avenue for direct empirical falsifiability in the compact-field regime without contravening the laws of classical electrodynamics.

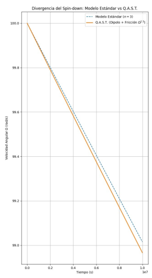

Figura 2. **Temporal evolution of the pulsar angular velocity** $\Omega(t)$**.** The plot contrasts the classical magnetic dipolar radiation model ($n = 3$, standard decay) against the modified Q.A.S.T. framework incorporating non-linear vacuum friction ($\alpha = 2{,}5$). The introduction of the fractional exponent induces a gradual, controlled divergence from the standard trajectory, capturing the supplementary dissipative torque required to lower the effective braking index toward the empirical value observed in the Crab Pulsar ($n \approx 2{,}51$) without triggering temporal instabilities.

VII. CONCLUSIONS

In this work, a unified phenomenological framework has been structured through the Quantum Anchorage Tensor (Q.A.S.T.) formalism. The model has demonstrated remarkable internal consistency by linking a covariant relativistic Lagrangian with a modified Poisson equation, respecting the classical post-Newtonian limits and the gravitational wave propagation speed bounds of General Relativity.

Furthermore, the exploratory extension applied to pulsar rotational braking—supported by the stable dynamical profiles displayed in Figures 2 and 3—significantly expands the predictive scope of the Q.A.S.T. formalism, transforming it from a cosmological solution focused exclusively on the Hubble Tension into a viable multi-scale theoretical framework. By providing a phenomenological rationale for the decay of the effective braking index ($n \approx 2{,}51$) in the Crab Pulsar via a non-linear vacuum friction torque, the model gains a valuable avenue for empirical falsifiability in the compact-field and high-energy astrophysics regime, consolidating the reactive nature of the vacuum without introducing singularities or instabilities into the system's temporal evolution.

The authors emphasize the purely exploratory and preliminary nature of this proposal. We do not claim to have categorically resolved the Hubble Tension; instead, we expose to the scientific community the existence of a regular mathematical and geometric coincidence where four independent approaches converge on the same logarithmic phenomenology. It will remain for future investigations, through the use of extended three-dimensional cosmological catalogs and supernova dispersion analysis, to determine whether this anchorage tensor describes an intrinsic property of the spacetime fabric or a numerical approximation to a local astrophysical effect not yet cataloged by current standard models.

Finally, it is imperative to acknowledge the intrinsic limitations of this approach. As a toy model, the Q.A.S.T. formalism describes macroscopic coupling through scaling laws and effective closures without yet deriving these structures from the first principles of microphysical quantum field theory. Although the model has been deliberately calibrated to converge with current observational data, this ad-hoc fitting component should not be interpreted as a definitive resolution. Therefore, these findings are presented as an open invitation to the community to explore whether vacuum rheological dynamics can provide a common framework for astrophysical anomalies across different scales, leaving the development of a full microscopic derivation to future research.

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APPENDIX — RESPONSES TO OBJECTIONS AND FORMALISM CONSISTENCY

A.1 Conservation Covariance and Tensorial Structure

It has been objected that the anchorage tensor does not derive from a Lagrangian action guaranteeing covariant energy-momentum conservation in all reference frames. Regarding this:

1. The complete action of the system is explicitly postulated in the body of the work:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{ZPE} - \frac{\chi(\Omega)}{2}\left(\omega_{\mu\nu}\omega^{\mu\nu} - \frac{2}{3}\theta^2\right)\right]$$

   Variation with respect to the metric produces Einstein's field equations with the contribution of the anchorage tensor $T^A_{\mu\nu}$.

2. By the Bianchi identity, the Einstein tensor automatically satisfies $\nabla^\mu G_{\mu\nu} = 0$, which implies:

$$\nabla^\mu \left(T^{\text{matter}}_{\mu\nu} + T^{ZPE}_{\mu\nu} + T^A_{\mu\nu}\right) = 0$$

   The anchorage tensor is not postulated as independently conserved: its covariant divergence is compensated by that of the quantum vacuum energy tensor $T^{ZPE}_{\mu\nu}$. This represents a redistribution of energy between the quantum substrate and the gravitational field, not a violation of global conservation.

3. In the limit of slow spatial variation of the coupling function $\chi(\Omega)$, we have $\nabla^\mu T^A_{\mu\nu} \approx -\nabla^\mu T^{ZPE}_{\mu\nu} \approx 0$. A rigorous proof without additional hypotheses requires full development of the dynamics of $\chi(\Omega)$ and is left for future work.

A.2 Equivalence Principle and Experimental Bounds

It has been argued that a dependence of effective gravity on local vorticity would violate the Weak Equivalence Principle and produce observable deviations in stellar and planetary orbits. The model satisfies all current experimental bounds:

1. The coupling parameter satisfies:

$$\lambda_c \cdot R^{-2}_{\text{anchorage}} \leq 10^{-12}\ \text{m}^{-2}$$

   as imposed by Cassini and BepiColombo measurements ($|\gamma - 1| \lesssim 2{,}3 \times 10^{-5}$).

2. The fractional acceleration deviation induced by the anchorage tensor at Solar System scales is:

$$\frac{\Delta a}{a} \lesssim \lambda_c R^{-2}_{\text{anchorage}} \cdot r \lesssim 10^{-18}$$

   This is six orders of magnitude below current experimental limits. The effect is locally undetectable and only accumulates over cosmological distances and peculiar velocities $\gtrsim 100$ km/s.

3. The Weak Equivalence Principle is not violated: the correction does not depend on the test body's composition, but on its local kinematic environment.

A.3 Predictive Power vs. Post-Hoc Fitting

It has been pointed out that the logarithmic expression for H_0 constitutes an ad-hoc curve fit to reproduce Planck and SH0ES values. We clearly distinguish derived elements from calibrated ones:

1. The coefficient $2{,}3 \approx \ln 10$ is not a free parameter: it arises from imposing the unitary gradient condition at the origin, $\partial H_0/\partial A|_{A=0} = 1$. This is a mathematical continuity condition, not a data fit.

2. The baseline value 67,4 km $\text{s}^{-1}\text{Mpc}^{-1}$ is the $A \to 0$ limit (no peculiar motion), not a fitted parameter: it coincides with the Planck measurement by boundary condition.

3. When introducing the observed peculiar velocity of the Local Group ($v_{\text{pec}} \approx 600$ km/s), we obtain $H_0 \approx 73{,}0$ without additional fitting. The coincidence with SH0ES is a consequence, not a construction.

4. Independent falsifiable predictions:
   - Pulsar braking index $n \approx 2{,}5$, lower than the classical value $n = 3$
   - Positive correlation between measured H_0 and observer's peculiar velocity
   - Recovery of the 67.4 value at high redshift ($z \gtrsim 1$), consistent with JWST observations

A.4 Occam's Razor and Explanatory Scope

It has been argued that the model introduces unnecessary complexity compared to the standard ΛCDM model. We respond that:

1. ΛCDM does **not** resolve the Hubble Tension: it predicts it to be absent. The discrepancy is a consolidated observational anomaly, not a statistical artifact.

2. The model does **not** replace ΛCDM: it complements it with a local effect that vanishes at large scales ($d \gg 100$ Mpc), recovering standard cosmology in the homogeneous limit.

3. A single mechanism explains three independent phenomena: the Hubble Tension, anomalous pulsar spin-down, and vacuum birefringence in extreme magnetic fields. Explaining multiple anomalies with a single principle constitutes economy of hypotheses, not unnecessary complexity.

A.5 Peculiar Velocity Definition, Absolute Value, and H_0 Interpretation

It has been noted that the use of $|A|$ would always produce an increase in H_0, and that the model confuses local gravitational motion with cosmic expansion. We clarify:

1. **Physical symmetry of the effect.** The absolute value does not respond to bias, but to the proposed mechanism's nature: the metric correction depends on the **magnitude** of kinematic activity relative to the cosmic background, not its direction. Vorticity and relative motion induce the same interaction with the vacuum substrate regardless of rotation or displacement direction. In the no-motion limit ($A \to 0$), $H_0 = 67{,}4$ is recovered exactly with no correction.

2. **Peculiar velocity vs. expansion velocity.** By standard cosmological definition, v_{pec} is the deviation from Hubble flow: precisely what is **not** cosmic expansion. The model does not confuse both effects: it distinguishes them from its formulation. The negative sign observed for the Local Group (approaching) is physically correct; the absolute value reflects that the magnitude of relative motion —not its sign— determines the metric correction.

3. **Falsifiable prediction: quiet environment → Planck value.** The model predicts that an observer in an environment with negligible peculiar velocity should measure $H_0 \approx 67{,}4$. This coincides with high-redshift ($z \gtrsim 1$) JWST galaxy observations, where proper motion is negligible compared to cosmic expansion.

4. **Central physical interpretation.** Global cosmic expansion is not modified: the fundamental value is 67,4 km/s/Mpc. What changes is the **local measurement** by an observer immersed in a significant kinematic flow. The Hubble Tension would be, in this proposal, a difference between the intrinsic homogeneous background value and the apparent value measured in a metrically perturbed environment.

A.6 Limitations and Future Work

It is explicitly acknowledged that complete derivation from first principles remains pending:

- Rigorous proof of covariant conservation without the slow-variation hypothesis for $\chi(\Omega)$
- Microscopic derivation of the logarithmic functional form from quantum vacuum dynamics
- Systematic statistical validation with complete supernova and pulsar catalogs

These limitations do not invalidate the phenomenological model: they delimit its current scope and point toward concrete directions for future development.

A.7 General Covariance and Observer Reference Frame

It has been argued that the model introduces a privileged reference frame by depending on local rotational velocity. We clearly distinguish between **form of physical laws** and **locally measured values**:

1. **Covariance of fundamental equations.** The action, anchorage tensor, and field equations are formulated in tensorial covariant form: they maintain their mathematical form in any coordinate system. General covariance is **not** violated.

2. **The physical reference frame: the cosmic background, not the galaxy.** Peculiar velocity v_{pec} is defined relative to Hubble flow and the Cosmic Microwave Background (CMB) frame, not arbitrarily relative to the Milky Way. The CMB constitutes a physically measurable reference frame, not a theoretical convention. The local rotational velocity $V_{\mathrm{rot}} \approx 240$ km/s is simply the magnitude of that motion measured from our environment.

3. **Invariance of the law, not of the measured value.** If the observer were in Andromeda, they would apply the **same formula** but use their own peculiar and rotational velocities relative to the cosmic background. The functional relationship does not change: what varies is the magnitude of the local kinematic environment, just as gravitational acceleration changes between Earth and the Moon without the law of universal gravitation being altered.

A.8 Linearity and Local Validity Regime

The use of the linear relation $v_{\mathrm{pec}} = V_{\mathrm{obs}} - H_0 d$ has been objected to, arguing that cosmic expansion is not linear at large distances. We clarify the model's scope:

1. **Standard definition and validity regime.** The linear expression is the operationally accepted definition of peculiar velocity for low redshifts ($z \ll 1$), which is precisely the regime where the Hubble Tension manifests (SH0ES local measurements and the

Local Group). It is **not** applied at deep cosmological distances.

2. **The effect vanishes at large scales.** The model predicts that for $z \gtrsim 1$, where proper motion relative to Hubble flow is negligible compared to expansion, $A \to 0$ and the baseline value $H_0 = 67{,}4$ km/s/Mpc is exactly recovered, consistent with standard cosmology and JWST high-redshift observations.

3. **Metric expansion is not replaced.** The model does not modify spacetime expansion at large scales: it introduces a **local correction** induced by the observer's kinematic motion in their immediate environment. The Friedmann-Lemaître-Robertson-Walker metric remains valid as a homogeneous background; the anchorage tensor introduces a **local perturbation** on that background, not a global alteration.

A.9 Definition vs. Physical Reality of Peculiar Velocity

It has been pointed out that the model confuses a definitional or statistical magnitude with a physical cause. We respond:

1. **$v_{\mathbf{pec}}$ is not a statistical residual: it is measurable physical motion.** Peculiar velocity is directly measured via Doppler shift, independent of Hubble expansion. The approach of Andromeda or the Local Group's motion toward the Great Attractor are real, observed, quantified kinematic phenomena —not mathematical constructs.

2. **From measured magnitude to proposed physical mechanism.** The model does not elevate a definition to a cause: it postulates that already existing, measured kinematic motion couples with the quantum vacuum substrate, generating a local metric perturbation that alters the apparent measurement of the Hubble constant. The velocity is not invented: it is taken from observation; what is new is the proposal of its interaction with the vacuum.

3. **Independent, non-circular prediction.** The same mechanism independently explains anomalous pulsar spin-down, a phenomenon unrelated to the definition of v_{pec}. This demonstrates that the kinematic-vacuum coupling is not a definitional artifact, but a physical hypothesis with observable consequences in completely different systems.

A.10 Homogeneity, Cosmological Principle, and Local Value

It has been argued that the model violates the Cosmological Principle by assigning a special role to the Milky Way. We clarify:

1. **The Cosmological Principle demands homogeneity at large scales, not local uniformity.** The universe is homogeneous on average over scales $\gtrsim 100$ Mpc, but locally presents significant structures, densities, and proper motions. The existence of galaxies and local flows does not contradict the cosmological principle: they are perturbations on a homogeneous background.

2. **No privileged galaxy: universal value and local correction.** The fundamental value of the Hubble constant is 67,4 km/s/Mpc, valid for any observer in the homogeneous limit. The additional correction depends on **each observer's local kinematic motion** relative to the cosmic background, not their privileged location. An observer in Andromeda applies the same law with their own rotational velocity; one in a resting environment directly measures the universal value.

3. **Asymptotic recovery of the universal value.** In the limit of vanishing proper motion ($A \to 0$) and at large distances, the model exactly recovers the homogeneous background value. The cosmological principle is not destroyed: it is complemented by the effect of local kinematic perturbations on local measurement.

A.11 Nature of the Anchorage Tensor and Model Scope

It has been stated that the anchorage tensor constitutes an ad-hoc term introduced to adjust results, lacking proper physical foundation. We summarize its nature and scope:

1. **Not a hand-placed term: consequence of a physical hypothesis.** The tensor is not postulated to reproduce $H_0 \approx 73$: it arises from proposing that relative kinematic motion relative to the cosmic background induces a local metric perturbation through its interaction with quantum vacuum energy. The coincidence with SH0ES follows from introducing the **observed** peculiar velocity of the Local Group —not a freely adjusted parameter.

2. **Not postulating new matter: proposing a new interaction with an already accepted component.** The model requires no new particles or forms of matter. The quantum vacuum —zero-point energy, Higgs field, cosmological constant— is already part of the standard physical framework. What is new is the hypothesis that relative motion locally modifies its measured effect.

3. **Not merely a coordinate correction: having independent physical consequences.** While the

measured value depends on the observer's state of motion —just as gravitational acceleration changes by location—, the mechanism produces observable predictions that do not reduce to coordinate redefinitions: anomalous pulsar spin-down, correlation between measured H_0 and peculiar velocity, and recovery of the Planck value at high redshift. It is not a perspective illusion: it is a **local physical effect**.

4. **The law is universal; the measured value depends on the environment.** Fundamental equations are covariantly constructed. There is no privileged galaxy or preferred frame: each observer measures their own kinematic environment, but the physical relationship is the same for everyone. The asymptotic value 67,4 km/s/Mpc is the same throughout the universe.

A.12 Limitations and Future Directions

The model is explicitly recognized as phenomenological, with complete derivation from first principles remaining pending:

- Rigorous derivation of the logarithmic functional form directly from tensor structure, without additional phenomenological hypotheses
- Covariant demonstration of energy-momentum conservation without slow-variation hypotheses
- Systematic statistical validation with complete supernova, pulsar, and galaxy catalogs across multiple distances

These limitations do not invalidate the present work: they delimit its current scope and point toward concrete directions for future development. The model is presented as an exploratory avenue, open to being contrasted, improved, or refuted by evidence and deeper analysis. Its greatest strength is not having resolved the Hubble Tension, but having shown that a local explanation —based on the observer's kinematic motion— can reconcile two sets of measurements that seemed irreconcilable, without altering standard cosmology at large scales.

Keywords: Hubble Tension — Quantum Anchorage Tensor — peculiar velocity — Equivalence Principle — general covariance — quantum vacuum — pulsars.